TeleCon firmware sketches

Which `.ino` to flash

ESP32 DevKit over Bluetooth, and ESP32-CAM for video

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Abstract

The Android app downloads one ZIP per connection from the firmware repository. Each ZIP is one Arduino sketch, and the sketch is the `.ino` with the same name. This note names every Bluetooth control sketch and every ESP32-CAM sketch, and the two ways to add a camera.

DevKit + Bluetooth CAM video only + DevKit

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1 Two ways to add a camera

Sticks, buttons, and telemetry always need a control sketch. Video is optional and uses an ESP32-CAM. Pick one of these layouts.

Layout	Boards	Sketches
Bluetooth only	One ESP32 DevKit	One of the three Bluetooth `.ino` files
Camera only, plus Bluetooth	ESP32-CAM for video, DevKit for sticks	`TeleCon_CAM_SoftAP_Video.ino` on the CAM, and one Bluetooth sketch on the DevKit

The phone joins the camera SoftAP for the picture. It does not send sticks over that video link. Sticks go to the control sketch: Bluetooth on the DevKit, or TCP on the single CAM that also serves video.

2 Bluetooth control on an ESP32 DevKit

Board in the app: **DevKit**. No camera on this board. Open the sketch folder in the Arduino IDE (the folder that contains the `.ino`) and upload to an ESP32 Dev Module.

ZIP / `.ino`	Phone setting	On the air
`TeleCon_Classic_Simpl`	Classic + Simple	Name `ESP32-TC-RC-BT-Simple`, PIN `1234`. Handshake `proto,simple`. Text lines `RC:CTRL`, `RC:BTN`, `RC:DATA`, `RC:PLOT`.
`TeleCon_Classic_Binar`	Classic + Binary	Name `ESP32-TC-RC-BT-Binary`. Handshake `proto,binary`. Frames `AA 55`, `BB 66`, `CC 11/22/33`.
`TeleCon_BLE_Binary`	BLE Binary	Name `ESP32-TC-RC-BLE-Binary` (Nordic UART). Same binary frames as Classic Binary.

`TeleCon_Classic_Simple.ino` Single file for a first board. Put motor and servo code in `onControl()` and `onButton()` where the file says `YOUR CODE HERE`. A steering servo on GPIO 13 is already wired in the sketch: stick `rx` = 0 is a 1500 µs pulse. Eight telemetry bytes (phone CH1–CH8) are chosen in `kTelemetryChannels[]` in this same file. With `ENABLE_PLOT` at 1 the board sends `RC:PLOT` about every 50 ms.

`TeleCon_Classic_Binary.ino` The `.ino` only starts Bluetooth and the vehicle modules. Pins, PWM, and the eight telemetry channels live in the other files in that folder. Edit `pins.h` for your driver and `TelemetryChannels.h` for what CH1–CH8 report. The phone setting must be Classic + Binary before connect. The wire format is the control protocol document, not the text lines of the Simple sketch.

`TeleCon_BLE_Binary.ino` Same vehicle modules and the same `TelemetryChannels.h` table as Classic Binary. The link is BLE instead of Classic SPP. Choose BLE Binary in the app. Do not pair it as a Classic device.

These three sketches are also the control half of the two-board camera layout below. Flash only one of them on the DevKit.

3 ESP32-CAM, video only

Use this when a DevKit already runs one of the Bluetooth sketches and you want a picture from a second board.

`TeleCon_CAM_SoftAP_Video.ino` ZIP `TeleCon_CAM_SoftAP_Video.zip`. Flash an ESP32-CAM (AI-Thinker). This sketch serves video and does not accept sticks.

- SoftAP `TeleCon-RC-CAM-Starter`, password `telecon1234`, address `192.168.4.1`.
- Picture: `http://192.168.4.1/stream` (MJPEG) or `/capture` (one JPEG).
- Stream quality: `http://192.168.4.1/camconfig`.
- No control port. There is no `RC:CONNECT` on this board.

On the phone, turn on the camera overlay and keep the Bluetooth connection pointed at the DevKit (Classic Simple, Classic Binary, or BLE Binary). The phone uses Wi-Fi for the CAM and Bluetooth for the sticks at the same time. Do not put stick control on the ESP32-CAM.

4 What the ZIP contains

The app downloads `zips/TeleCon_<name>.zip`. Unzip it and open the folder that contains the `.ino` of the same name. Arduino compiles every `.cpp` and `.h` in that folder, so a binary sketch is the `.ino` plus those modules. Do not copy `SteerCenter` back in: stock sketches do not store a steering center. Trim stays on the phone, and `rx` $= 0$ is 1500 µs.